

Mutually Independent Hamiltonian Paths in Star Networks

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Abstract

Let $P_1 = \langle v_1, v_2, \dots, v_k \rangle$ and $P_2 = \langle u_1, u_2, \dots, u_k \rangle$ be any two hamiltonian paths of G . We say that P_1 and P_2 are *independent* if $u_1 = v_1$, $v_k = u_k$, and $v_i \neq u_i$ for $1 < i < k$. We say a set of hamiltonian paths $P_1, P_2, \dots, P_s$ of G are *mutually independent* if any two different hamiltonian paths are independent. We prove that there exist $(n - 2)$ mutually independent hamiltonian paths in S_n between any two nodes from different bipartite sets if $n \geq 4$. Moreover, this result is optimal.

Keywords: hamiltonian, hamiltonian laceable.

1 Introduction

For definitions and notations we follow [2]. $G = (V, E)$ is a *graph* if V is a finite set and E is a subset of $\{(u, v) \mid (u, v) \text{ is an unordered pair of } V\}$. We say that V is the *node set* and E is the *edge set*. For a node u , $N(u)$ denotes the *neighborhood* of u which is the set $\{v \mid (u, v) \in E\}$. For any node x of V , denote the *degree* of x by $\deg_G(x) = |N(x)|$. Two nodes u and v are *adjacent* if $(u, v) \in E$. A *path* is represented by $\langle v_1, v_2, \dots, v_k \rangle$ where $v_i \neq v_j$ for $i, j \in \{1, 2, \dots, k\}$ and $i \neq j$ and $(v_i, v_{i+1}) \in E$. Denote by $Q(i)$ the i -th node v_i of path $Q = \langle v_1, v_2, \dots, v_k \rangle$. We also write the path $\langle v_1, v_2, \dots, v_k \rangle$ as $\langle v_1, Q_1, v_i, v_{i+1}, \dots, v_j, Q_2, v_t, \dots, v_k \rangle$, where Q_1 is the path $\langle v_1, v_2, \dots, v_i \rangle$ and Q_2 is the path

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$\langle v_j, v_{j+1}, \dots, v_t \rangle$. We use $d(u, v)$ to denote the *distance* between u and v , i.e., the length of the shortest path joining u and v .

A path P is a *hamiltonian* path if $V(P) = V(G)$. Let $P_1 = \langle v_1, v_2, \dots, v_k \rangle$ and $P_2 = \langle u_1, u_2, \dots, u_k \rangle$ be two paths of G . We say that P_1 and P_2 are *independent* if $u_1 = v_1$, $v_k = u_k$, and $v_i \neq u_i$ for $1 < i < k$. We say a set of hamiltonian paths $P_1, P_2, \dots, P_s$ of G are *mutually independent* if any two different paths in the set are independent.

An interconnection network connects the processors of the parallel computer. Its architecture can be represented as a graph in which the nodes corresponds to processors and the edges to connections. Hence we use graphs and networks interchangeably. There are many mutually conflicting requirements in designing the topology for computer networks. The n -cube is one of the most popular topologies [7]. The star network S_n was proposed in [1] as "an attractive alternative to the n -cube" topology for interconnecting processors in parallel computers.

Since its introduction, the network received considerable attention. Some interesting properties of star networks are studied in [3, 4, 5, 6, 8]. For example, it is known that the star network is a bipartite graph and there is a hamiltonian path joining nodes from different partite sets. In particular, Fragopoulou and Akl [4, 5] study the embedding of $(n - 1)$ directed edge-disjoint spanning trees on the star network S_n . These spanning trees are used in communication algorithms for star networks. In this paper, we prove that there exist $(n - 2)$ mutually independent hamiltonian paths of S_n between any two nodes from different bipartite sets. Obviously, these mutually independent paths can be used for communication algorithms for star networks.

In the following section, we give the definition of the star networks and some of the previous work that are used in this paper. In section 3, we prove that there exist $(n - 2)$ mutually independent hamiltonian paths of S_n between any two nodes from different bipartite sets. Moreover, we prove that the value $(n - 2)$ is tight.

2 Basic properties of the star networks

Let n be a positive integer. We use $\langle n \rangle$ to denote the set $\{1, 2, \dots, n\}$. The n -dimensional star network, denoted by S_n , is a graph with the node set $V(S_n) = \{u_1 u_2 \dots u_n \mid u_i \in \langle n \rangle \text{ and } u_i \neq u_j \text{ for } i \neq j\}$. The edges are specified as follows: $u_1 u_2 \dots u_i \dots u_n$ is adjacent to $v_1 v_2 \dots v_i \dots v_n$ through an edge in dimension i with $2 \leq i \leq n$ if $v_j = u_j$ for $j \notin \{1, i\}$, $v_1 = u_i$ and $v_i = u_1$. By definition, S_n is an $(n - 1)$ -regular graph with $n!$ nodes. Moreover, it is node transitive and edge transitive.

We use bold face letters to denote nodes in S_n . Hence $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n$ is a sequence of nodes in S_n . It is known that the connectivity of S_n is $n - 1$. We use $\mathbf{e}$ to denote the element $12 \dots n$. It is known that S_n is a bipartite graph with one partite set containing those nodes corresponding to odd permutations and the other partite set containing those nodes

corresponding to even permutations. We will use white nodes to represent nodes for even permutations and black nodes to represent nodes for odd permutations. Let $\mathbf{u} = u_1 u_2 \dots u_n$ be any node of the star network S_n . We say that u_i is the i -th coordinate of $\mathbf{u}$, denoted by $(\mathbf{u})_i$, for $1 \leq i \leq n$. By the definition of S_n , there is exactly one neighbor $\mathbf{v}$ of $\mathbf{u}$ such that $\mathbf{u}$ and $\mathbf{v}$ are adjacent through an edge in the i -th dimension, $2 \leq i \leq n$. For this reason, we use $(\mathbf{u})^i$ to denote the unique i -neighbor of $\mathbf{u}$. Obviously, $((\mathbf{u})^i)^i = \mathbf{u}$. For $1 \leq i \leq n$, let S_n^i denote the subgraph of S_n induced by those nodes $\mathbf{u}$ with $(\mathbf{u})_n = i$. Obviously, S_n can be decomposed into n subgraphs S_n^i , $1 \leq i \leq n$, and each S_n^i is isomorphic to S_{n-1} . This furnishes a recursive definition (construction) for star networks. For $1 \leq i \neq j \leq n$, we use $E^{i,j}$ to denote the set of edges between S_n^i and S_n^j .

Lemma 1 $|E^{i,j}| = (n-2)!$ for any $1 \leq i \neq j \leq n$. Moreover, there are $\frac{(n-2)!}{2}$ edges joining black nodes of S_n^i to white nodes of S_n^j .

Proof. Let $i \neq j$. For each of $(n-2)!$ permutations of $\langle n \rangle - \{i, j\}$ there is exactly one transposition that represents an edge between a node of S_n^i and a fixed node of S_n^j . Hence $|E^{i,j}| = (n-2)!$. Exactly half of these correspond to transpositions of an odd permutation in S_n^i to an even permutation in S_n^j . $\square$

Lemma 2 Let $\mathbf{u}$ and $\mathbf{v}$ be two distinct nodes of S_n with $(\mathbf{u})_n = (\mathbf{v})_n$ such that $1 \leq d(\mathbf{u}, \mathbf{v}) \leq 2$. Then $((\mathbf{u})^n)_n \neq ((\mathbf{v})^n)_n$.

Theorem 1 [8] S_n is hamiltonian laceable if and only if $n \neq 3$.

Theorem 2 Assume that $n \geq 5$. Let $\{a_1, a_2, \dots, a_r\}$ be a subset of $\langle n \rangle$ for some $r \leq n$. For any white node $\mathbf{u}$ of $S_n^{a_1}$ and any black node $\mathbf{v}$ of $S_n^{a_r}$, there exists a path P as $\langle \mathbf{x}_1, P_1, \mathbf{y}_1, \mathbf{x}_2, P_2, \mathbf{y}_2, \dots, \mathbf{x}_r, P_r, \mathbf{y}_r \rangle$ joining $\mathbf{u}$ and $\mathbf{v}$ such that $\mathbf{x}_1 = \mathbf{u}$, $\mathbf{y}_r = \mathbf{v}$, and P_i is a hamiltonian path of $S_n^{a_i}$ joining $\mathbf{x}_i$ to $\mathbf{y}_i$ for $1 \leq i \leq r$. Moreover, P is a hamiltonian path of the subgraph induced by $\cup_{i=1}^r S_n^{a_i}$ joining $\mathbf{u}$ to $\mathbf{v}$.

Proof. We set that $\mathbf{x}_1$ as $\mathbf{u}$ and set that $\mathbf{y}_r$ as $\mathbf{v}$. Since $S_n^{a_1}$ is isomorphic to S_{n-1} , by Theorem 1, this theorem holds for $r = 1$. Assume that $r \geq 2$. By Lemma 1, we choose $(\mathbf{y}_i, \mathbf{x}_{i+1}) \in E^{a_i, a_{i+1}}$ with $\mathbf{y}_i$ is a black node and $\mathbf{x}_{i+1}$ is a white node for $1 \leq i \leq r-1$. Obviously, $\mathbf{x}_i \neq \mathbf{y}_i$ in $S_n^{a_i}$. Since $S_n^{a_i}$ is isomorphic to S_{n-1} , by Theorem 1, there is a hamiltonian path P_i of $S_n^{a_i}$ joining $\mathbf{x}_i$ to $\mathbf{y}_i$. Obviously, $\langle \mathbf{x}_1, P_1, \mathbf{y}_1, \mathbf{x}_2, P_2, \mathbf{y}_2, \dots, \mathbf{x}_r, P_r, \mathbf{y}_r \rangle$ is the required path. $\square$

Lemma 3 Let $\mathbf{w}$ be any black node and $\mathbf{u}, \mathbf{v}$ be any two distinct white nodes of S_n with $n \geq 4$. Then there exists a hamiltonian path in $S_n - \{\mathbf{w}\}$ joining $\mathbf{u}$ to $\mathbf{v}$.

Proof. We prove this lemma by induction. For $n = 4$, we can set $\mathbf{w} = 1234$ because S_4 is node transitive. The corresponding hamiltonian path between $\mathbf{u}$ and $\mathbf{v}$ in $S_4 - \{\mathbf{w}\}$ are listed below.

Then $\langle \mathbf{u}, P, \mathbf{s}, (\mathbf{s})^n, Q, \mathbf{t}, (\mathbf{t})^n, R, \mathbf{v} \rangle$ forms the desired path.

Case 2. $\mathbf{v} \in S_n^{n-1}$. Let $\{a_1, a_2, \dots, a_{n-2}\} = \langle n-1 \rangle - \{n-1\}$ with $a_1 = n$. Since S_n^i is isomorphic to S_{n-1} with $i \in \{a_1, a_2, \dots, a_{n-2}\}$, by Theorem 2, there exists a hamiltonian path R of the subgraph induced by $\cup_{i=1}^{n-1} S_n^{a_i}$ joining $\mathbf{u}$ to a black node $\mathbf{s}$ with $(\mathbf{s})_n = n-1$ and $(\mathbf{s})^n \neq \mathbf{v}$. Since S_n^{n-1} is isomorphic to S_{n-1} , by induction, there exists a hamiltonian path Q of $S_n^{n-1} - \{\mathbf{w}\}$ joining the white node $(\mathbf{s})^n$ to $\mathbf{v}$. Then $\langle \mathbf{u}, R, \mathbf{s}, (\mathbf{s})^n, Q, \mathbf{v} \rangle$ forms the desired path.

Case 3. $\mathbf{v} \in S_n^n$. Note that $|N(\mathbf{v}) - (\mathbf{v})^n| \geq 3$. We can choose $\mathbf{x}$ as a neighbor of $\mathbf{v}$ with $(\mathbf{x})_n \neq n-1$. Since S_n^n is isomorphic to S_{n-1} , there exists a hamiltonian path P of $S_n^n - \{\mathbf{x}\}$ joining $\mathbf{u}$ to $\mathbf{v}$. Without loss generality, we can write P as $\langle \mathbf{u}, P', \mathbf{s}, \mathbf{u} \rangle$. Since $d(\mathbf{s}, \mathbf{x}) \leq 2$, by Lemma 2, $((\mathbf{s})^n)_n \neq ((\mathbf{x})^n)_n$. Obviously, $\mathbf{x}$ and $\mathbf{s}$ both are black nodes.

Assume that $(\mathbf{s})_n = n-1$. Let $\{a_1, a_2, \dots, a_{n-2}\} = \langle n-2 \rangle$ with $a_{n-2} = (\mathbf{x})_n$. Since S_n^{n-1} are isomorphic to S_{n-1} , by induction, there exists a hamiltonian path Q of $S_n^{n-1} - \{\mathbf{w}\}$ joining the white node $(\mathbf{s})^n$ to a white node $\mathbf{t}$ with $(\mathbf{t})_n = a_1$. By Theorem 2, there exists a hamiltonian path R of the subgraph induced by $\cup_{i=1}^r S_n^{a_i}$ joining the black node $(\mathbf{t})^n$ to the white node $(\mathbf{x})^n$. Then $\langle \mathbf{u}, P', \mathbf{s}, (\mathbf{s})^n, Q, \mathbf{t}, (\mathbf{t})^n, R, (\mathbf{x})^n, \mathbf{b}, \mathbf{v} \rangle$ forms the desired path.

Assume that $(\mathbf{s})_n \neq n-1$. Let $\{a_1, a_2, \dots, a_{n-2}\} = \langle n-2 \rangle$ with $a_{n-3} = (\mathbf{y})_n$ and $a_{n-2} = (\mathbf{s})_n$. Since $S_n^{a_{n-2}}$ is isomorphic to S_{n-1} , by Theorem 1, there exists a hamiltonian path Q of $S_n^{a_{n-2}}$ joining the white node $(\mathbf{s})^n$ to a black node $\mathbf{t}$ with $(\mathbf{t})_n = n-1$. Since S_n^{n-1} is isomorphic to S_{n-1} , by induction, there exists a hamiltonian path H of $S_n^{n-1} - \{\mathbf{w}\}$ joining the white node $(\mathbf{t})^n$ to a white node $\mathbf{z}$ with $(\mathbf{z})_n = a_1$. By Theorem 2, there exists a hamiltonian path R of the subgraph induced by $\cup_{i=1}^r S_n^{a_i}$ joining the black node $(\mathbf{z})^n$ to the white node $(\mathbf{x})^n$. Then $\langle \mathbf{u}, P', \mathbf{s}, (\mathbf{s})^n, Q, \mathbf{t}, (\mathbf{t})^n, H, \mathbf{z}, (\mathbf{z})^n, R, (\mathbf{x})^n, \mathbf{x}, \mathbf{v} \rangle$ form the desired path. $\square$

Theorem 3 Let $\mathbf{u}$ be any white node of S_n with $n \geq 4$ and $\{a_1, a_2, \dots, a_{n-1}\}$ be an $n-1$ subset of $\langle n \rangle$. Then there exist hamiltonian paths $P_1, P_2, \dots, P_{n-1}$ of S_n joining $\mathbf{u}$ to some black node $\mathbf{z}_i$ such that

- (1) $(\mathbf{z}_i)_1 = (P_i(n!))_1 = a_i$ for $1 \leq i \leq n-1$, and
- (2) $|\{P_1(i), P_2(i), \dots, P_{n-1}(i)\}| = n-1$ for $2 \leq i \leq n!$.

Proof. Since S_n is node transitive, we may assume that $\mathbf{u} = \mathbf{e}$. We prove the theorem holds on S_4 by exhibiting three required hamiltonian paths in S_4 as follows:

$\{a_1, a_2, a_3\} = \{1, 2, 3\}$
(1234)(2134)(3124)(1324)(2314)(3214)(4213)(1243)(3241)(4231)(2431)(3421)(4321)(2341)(1342)(4312)(3412)(2413)(1423)(4123)(2143)(3142)(4132)(1432)(1234)(3214)(2314)(4312)(1342)(3142)(4132)(1432)(3412)(2413)(4213)(1243)(2143)(4123)(1423)(3421)(2431)(4231)(3241)(2341)(4321)(1324)(3124)(2134)(1234)(4231)(2431)(1432)(3412)(4312)(1342)(3142)(4132)(1432)(4132)(2134)(3124)(1324)(2314)(3214)(4213)(1243)(3241)(4231)(2431)(3421)(4321)(2341)(1342)(4312)(3412)(2413)(1423)(4123)(2143)(3142)(4132)(1432)(1234)(2134)(3124)(1324)(2314)(3214)(4213)(2413)(1423)(4123)(2143)(1243)(3241)(4231)(2431)(3421)(4321)(2341)(1342)(3142)(4132)(1432)(3412)(4312)
$\{a_1, a_2, a_3\} = \{1, 2, 4\}$
(1234)(4231)(2431)(3421)(4321)(2341)(3241)(1243)(2143)(3142)(1342)(4312)(3412)(1432)(4132)(2134)(3124)(4123)(1423)(2413)(4213)(3214)(2314)(1324)(1234)(3214)(4213)(1243)(3241)(4231)(2431)(1432)(3412)(2413)(1423)(3421)(4321)(2341)(1342)(4312)(2314)(1324)(3124)(4123)(2143)(3142)(4132)(2143)(1234)(2134)(3124)(1324)(2314)(3214)(4213)(2413)(1423)(4123)(2143)(1243)(3241)(4231)(2431)(3421)(4321)(2341)(1342)(3142)(4132)(1432)(3412)(4312)
$\{a_1, a_2, a_3\} = \{2, 3, 4\}$
(1234)(3214)(4213)(1243)(3241)(4231)(2431)(1432)(3412)(2413)(1423)(3421)(4321)(2341)(1342)(4312)(2314)(1324)(3124)(4123)(2143)(3142)(4132)(2143)(1234)(4231)(2431)(1432)(3412)(4312)(1342)(3142)(4132)(2134)(3124)(1324)(2314)(3214)(4213)(2413)(1423)(4123)(2143)(1243)(3241)(4231)(2431)(3421)(4321)(2341)(1342)(3142)(4132)(1432)(3412)(4312)

Assume that the theorem holds for S_k , $4 \leq k < n$. Without loss of generality, we suppose that $a_1 < a_2 < \dots < a_{n-1}$. We set that $a'_i = a_i$ for $1 \leq i \leq n-3$, $a'_{n-2} = a_{n-1}$, and $a'_{n-1} = a_{n-2}$. Obviously, $(\mathbf{e})^n$ is a black node in S_n^1 . We set that

$$b_i = \begin{cases} i+3 & \text{if } 1 \leq i \leq n-4 \\ i-n+4 & \text{if } n-3 \leq i \leq 2n-5 \\ i+2n+5 & \text{if } 2n-4 \leq i \leq 2n-3 \end{cases}$$

By induction hypothesis, there exist hamiltonian paths $H_1, H_2, \dots, H_{n-2}$ of S_n^n joining $\mathbf{e}$ to some black node $\mathbf{v}_i$ such that

- (1) $(\mathbf{v}_i)_1 = (H_i((n-1)!))_1 = i+3$ for $1 \leq i \leq n-4$, $(\mathbf{v}_{n-3})_1 = (H_{n-3}((n-1)!))_1 = 1$, $(\mathbf{v}_{n-2})_1 = (H_{n-2}((n-1)!))_1 = 3$, and
- (2) $|\{H_1(i), H_2(i), \dots, H_{n-2}(i)\}| = n-2$ for $2 \leq i \leq (n-1)!$.

Since $|\{\mathbf{v} \in S_n^{i+2} \mid \mathbf{v} \text{ is a black node with } (\mathbf{v})_1 = a'_i\}| = \frac{(n-2)!}{2} \geq 2$, we choose a black node $\mathbf{z}_i \in S_n^{b_i+n-2}$ with $(\mathbf{z}_i)_1 = a'_i$ for $1 \leq i \leq n-3$. Since S_n^i is isomorphic to S_{n-1} for $1 \leq i \leq n$, by Theorem 2, there exists a path W_i as $\langle \mathbf{x}_1^i, T_1^i, \mathbf{y}_1^i, \mathbf{x}_2^i, T_2^i, \mathbf{y}_2^i, \dots, \mathbf{x}_{n-1}^i, T_{n-1}^i, \mathbf{y}_{n-1}^i \rangle$ joining the white node $(\mathbf{v}_i)^n$ to the black node $\mathbf{z}_i$ such that $\mathbf{x}_1^i = (\mathbf{v}_i)^n$, $\mathbf{y}_{n-2}^i = \mathbf{z}_i$, and T_j^i is a hamiltonian path of $S_n^{b_i+j-1}$ joining $\mathbf{x}_j^i$ to $\mathbf{y}_j^i$ for $1 \leq j \leq n-1$, and $1 \leq i \leq n-2$. Again, there exists a path W_{n-2} as $\langle \mathbf{x}_1^{n-2}, T_1^{n-2}, \mathbf{y}_1^{n-2}, \mathbf{x}_2^{n-2}, T_2^{n-2}, \mathbf{y}_2^{n-2}, \dots, \mathbf{x}_{n-1}^{n-2}, T_{n-1}^{n-2}, \mathbf{y}_{n-1}^{n-2} \rangle$ joining the white node $(\mathbf{v}_{n-2})^n$ to the black node $\mathbf{z}_{n-2}$ such that $\mathbf{x}_1^{n-2} = (\mathbf{v}_{n-2})^n$, $\mathbf{y}_{n-1}^{n-2} = \mathbf{z}_{n-2}$, and T_j^{n-2} is a hamiltonian path of $S_n^{b_i+n-2}$ joining $\mathbf{x}_j^{n-2}$ to $\mathbf{y}_j^{n-2}$ for $1 \leq j \leq n-1$. We set that P_i as $\langle \mathbf{e}, H_i, \mathbf{v}_i, (\mathbf{v}_i)^n, W_i, \mathbf{z}_i \rangle$ for $1 \leq i \leq n-2$.

By Theorem 2, there exists a path W as $\langle \mathbf{x}_1, Q_1, \mathbf{y}_1, \mathbf{x}_2, Q_2, \mathbf{y}_2, \dots, \mathbf{x}_{n-1}, Q_{n-1}, \mathbf{y}_{n-1} \rangle$ joining the black node $(\mathbf{e})^n$ to the white node $((\mathbf{e})^{n-1})^n$ such that $\mathbf{x}_1 = (\mathbf{e})^n$, $\mathbf{y}_{n-1} = ((\mathbf{e})^{n-1})^n$, and Q_i is a hamiltonian path of S_n^i joining $\mathbf{x}_i$ to $\mathbf{y}_i$ for $1 \leq i \leq n-1$. By Lemma 1, $|\{\mathbf{v} \in S_n^n \mid \mathbf{v} \text{ is a black node with } (\mathbf{v})_1 = a'_{n-1}\}| = \frac{(n-2)!}{2} \geq 2$, we choose a black node $\mathbf{z}_{n-1} \in S_n^1$ with $(\mathbf{z}_{n-1})_1 = a'_{n-1}$ and $\mathbf{z}_{n-1} \neq (\mathbf{e})^{n-1}$. Since S_n^n is isomorphic to S_{n-1} , by Lemma 3, there exists a hamiltonian path R of $S_n^n - \{\mathbf{e}\}$ joining the black node $(\mathbf{e})^{n-1}$ to $\mathbf{z}_{n-1}$. We set that P_{n-2} as $\langle \mathbf{e}, W, ((\mathbf{e})^{n-1})^n, (\mathbf{e})^{n-1}, R, \mathbf{z}_{n-1} \rangle$.

Then $\{P_1, P_2, \dots, P_{n-1}\}$ forms the desired paths.

We illustrate the proof of Theorem 3 with $n = 5$ as follows: Obviously, $a_1 \neq 3$, $a_2 \neq 4$, $a_3 \neq 5$, and $a_4 \neq 2$. By induction hypothesis, there exist hamiltonian paths H_1, H_2, H_3 of S_5^5 joining $\mathbf{e}$ to some black node $\mathbf{v}_i$ such that

- (1) $(\mathbf{v}_1)_1 = (H_1(24))_1 = 4$, $(\mathbf{v}_2)_1 = (H_2(24))_1 = 1$, $(\mathbf{v}_3)_1 = (H_3(24))_1 = 3$, and
- (2) $|\{H_1(i), H_2(i), H_3(i)\}| = 3$ for $2 \leq i \leq 24$.

Then we choose a black node $\mathbf{z}_1$ in S_5^3 with $(\mathbf{z}_1)_1 = a_1$, a black node $\mathbf{z}_2$ in S_5^4 with $(\mathbf{z}_2)_1 = a_2$, a black node $\mathbf{z}_3$ in S_5^2 with $(\mathbf{z}_3)_1 = a_4$, and a black node $\mathbf{z}_4$ in S_5^2 with $(\mathbf{z}_4)_1 = a_3$. By Theorem 2, there exists a path W_1 as $\langle \mathbf{x}_1^1, T_1^1, \mathbf{y}_1^1, \mathbf{x}_2^1, T_2^1, \mathbf{y}_2^1, \mathbf{x}_3^1, T_3^1, \mathbf{y}_3^1, \mathbf{x}_4^1, T_4^1, \mathbf{y}_4^1 \rangle$ joining

the white node $(\mathbf{v}_1)^5$ to the black node $\mathbf{z}_1$ such that $\mathbf{x}_1^1 = (\mathbf{v}_1)^5$, $\mathbf{y}_4^1 = \mathbf{z}_1$, and T_j^1 is a hamiltonian path of $S_n^{b_j}$ joining $\mathbf{x}_j^1$ to $\mathbf{y}_j^1$ for $1 \leq j \leq 4$, and a path W_2 . Furthermore, a path W_3 .

By Theorem 2, there exists a path W as $\langle \mathbf{x}_1, Q_1, \mathbf{y}_1, \mathbf{x}_2, Q_2, \mathbf{y}_2, \mathbf{x}_3, Q_3, \mathbf{y}_3, \mathbf{x}_4, Q_4, \mathbf{y}_4 \rangle$ joining the black node $(\mathbf{e})^5$ to the white node $((\mathbf{e})^4)^5$ such that $\mathbf{x}_1 = (\mathbf{e})^5$, $\mathbf{y}_4 = ((\mathbf{e})^4)^5$, and Q_i is a hamiltonian path of S_5^i joining $\mathbf{x}_i$ to $\mathbf{y}_i$ for $1 \leq i \leq 4$. Since S_5^5 is isomorphic to S_4 , by Lemma 3, there exists a hamiltonian path R of $S_5^5 - \{\mathbf{e}\}$ joining the black node $(\mathbf{e})^4$ to $\mathbf{z}_4$. We set that P_3 as $\langle \mathbf{e}, W, ((\mathbf{e})^4)^5, (\mathbf{e})^4, R, \mathbf{z}_4 \rangle$.

Thus $\{P_1, P_2, P_3, P_4\}$ forms the desired paths. See Figure 1 for illustration. $\square$

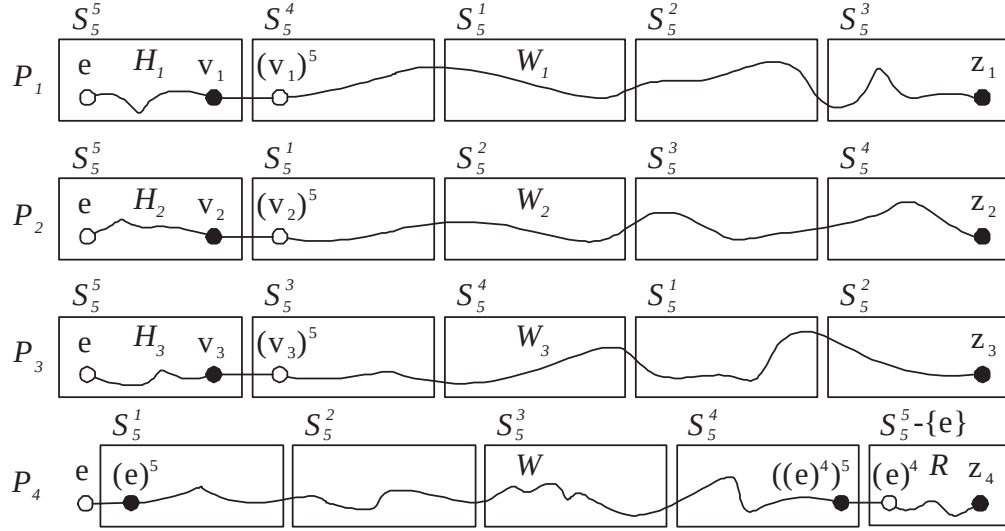


Figure 1: Illustration for Theorem 3 on S_5 .

3 Mutually independent hamiltonian paths

Theorem 4 Assume that $n \geq 4$. Then there exist $(n-2)$ mutually independent hamiltonian paths of S_n between any two nodes $\mathbf{u}$ and $\mathbf{v}$ from different bipartite sets. Moreover the value $(n-2)$ is tight.

Proof. Since star networks are node transitive, we may assume that $\mathbf{u} = \mathbf{e}$. We prove this theorem by induction. For $n \geq 4$, the required hamiltonian paths are list below:

$\mathbf{v} = (2134)$ (1234)(4231)(2431)(1432)(4132)(3142)(2143)(1243)(3241)(2341)(1342)(4312)(3412)(2413)(4213)(3214)(2314)(1324)(4321)(3421)(1423)(4123)(3124)(2134) (1234)(3214)(4213)(2413)(3412)(4312)(2314)(1324)(3124)(4123)(1423)(3421)(4321)(2341)(1342)(3142)(2143)(1243)(3241)(4231)(2431)(1432)(4132)(2134)
$\mathbf{v} = (3214)$ (1234)(2134)(3124)(1324)(2314)(4312)(3412)(1432)(4132)(3142)(1342)(2341)(4321)(3421)(2431)(4231)(3241)(1243)(2143)(4123)(1423)(2413)(4213)(3214) (1234)(4231)(2431)(3421)(4321)(2341)(3241)(1243)(4213)(2413)(1423)(4123)(2143)(3142)(1342)(4312)(3412)(1432)(4132)(2134)(3124)(1324)(2314)(3214)
$\mathbf{v} = (4231)$ (1234)(2134)(4132)(1432)(2431)(3421)(4321)(1324)(3124)(4123)(1423)(2413)(3412)(4312)(2314)(3214)(4213)(1243)(2143)(3142)(1342)(2341)(3241)(4231) (1234)(3214)(2314)(4312)(3412)(2413)(4213)(1243)(3241)(2341)(1342)(3142)(2143)(4123)(1423)(3421)(4321)(1324)(3124)(2134)(4132)(1432)(2431)(4231)
$\mathbf{v} = (1432)$ (1234)(2134)(4132)(3142)(1342)(2341)(4321)(1324)(3124)(4123)(2143)(1243)(3241)(4231)(2431)(3421)(1423)(2413)(4213)(3214)(2314)(4312)(3412)(1432) (1234)(3214)(2314)(4312)(3412)(2413)(4213)(1243)(3241)(4231)(2431)(3421)(1423)(4123)(2143)(3142)(1342)(4312)(3412)(1432)(4132)(2134)(3124)(1324)(2314)(3214)
$\mathbf{v} = (1243)$ (1234)(4231)(3241)(2341)(1342)(4312)(3412)(1432)(2431)(3421)(4321)(1324)(2314)(3214)(4213)(2413)(1423)(4123)(3124)(2134)(4132)(3142)(2143)(1243) (1234)(2134)(4132)(3142)(2143)(4123)(3124)(1324)(2314)(3214)(4213)(2413)(1423)(3421)(4321)(2341)(1342)(4312)(3412)(1432)(2431)(4231)(3241)(1243)
$\mathbf{v} = (1324)$ (1234)(2134)(3124)(4123)(1423)(2413)(3412)(1432)(4132)(3142)(2143)(1243)(4213)(3214)(2314)(4312)(1342)(2341)(3241)(4231)(2431)(3421)(4321)(1324) (1234)(4231)(2431)(3421)(4321)(2341)(3241)(1243)(4213)(3214)(2314)(4312)(1342)(3142)(2143)(4123)(1423)(2413)(3412)(1432)(4132)(2134)(3124)(1324)
$\mathbf{v} = (2341)$ (1234)(2134)(4132)(3142)(1342)(4312)(2314)(3214)(4213)(1243)(2143)(4123)(3124)(1324)(4321)(3421)(1423)(2413)(3412)(1432)(2431)(4231)(3241)(2341) (1234)(3214)(4213)(1243)(3241)(4231)(2431)(3421)(1423)(2413)(3412)(1432)(4132)(2134)(3124)(4123)(2143)(3142)(1342)(4312)(2314)(1324)(4321)(2341)
$\mathbf{v} = (3142)$ (1234)(3214)(2314)(4312)(1342)(2341)(3241)(4231)(2431)(1432)(3412)(2413)(4213)(1243)(2143)(4123)(1423)(3421)(4321)(1324)(3124)(2134)(4132)(3142) (1234)(4231)(2431)(1432)(4132)(2134)(3124)(4123)(1423)(3421)(4321)(1324)(2314)(4312)(3412)(2143)(3142)(4213)(1342)(2341)(3241)(1243)(2143)(3142)
$\mathbf{v} = (4123)$ (1234)(4231)(2431)(3421)(1423)(2413)(4213)(3214)(2314)(1324)(4321)(2341)(3241)(1243)(2143)(3142)(1342)(4312)(3412)(1432)(4132)(2134)(3124)(4123) (1234)(3214)(2314)(1324)(3124)(4132)(3412)(1342)(4312)(4321)(2341)(4321)(3241)(2341)(4321)(3421)(1423)(2413)(4213)(1243)(2143)(4123)
$\mathbf{v} = (4312)$ (1234)(4231)(3241)(2341)(1342)(3142)(4132)(2134)(3124)(1324)(4321)(3421)(2431)(1432)(3412)(2413)(1423)(4123)(2143)(1243)(4213)(3214)(2314)(4312) (1234)(2134)(3124)(1324)(2314)(3214)(4213)(2413)(1423)(4123)(2143)(1243)(3241)(4231)(2431)(3421)(4321)(2341)(1342)(3142)(4132)(1432)(3412)(4312)
$\mathbf{v} = (2413)$ (1234)(2134)(3124)(4123)(1423)(3421)(2431)(4231)(3241)(1243)(2143)(3142)(4132)(1432)(3412)(4312)(1342)(2341)(4321)(1324)(2314)(3214)(4213)(2413) (1234)(4231)(3241)(1243)(4213)(3214)(2314)(4312)(1342)(2341)(4321)(1324)(3124)(4132)(3142)(2143)(4123)(1423)(3421)(4321)(2431)(1432)(3412)(2413)
$\mathbf{v} = (3421)$ (1234)(3214)(4213)(2413)(1423)(4123)(3124)(2134)(4132)(1432)(3412)(4312)(2314)(1324)(4321)(2341)(1342)(3142)(2143)(1243)(3241)(4231)(2431)(3421) (1234)(2134)(4132)(1432)(2431)(4231)(3241)(2341)(1342)(3142)(2143)(1243)(4213)(3214)(2314)(4312)(3412)(2413)(1423)(4123)(3124)(1324)(4321)(3421)

Assume that the theorem holds for S_k , $4 \leq k < n$. Since S_n is node transitive and edge transitive, we assume that $\mathbf{u} = \mathbf{e} \in S_n^n$ and $\mathbf{v} \in S_n^{n-1}$. We set $b_i = i$ for $1 \leq i \leq n-2$ and $b_i = i - n + 2$ for $n-1 \leq i \leq 2n-5$. By Theorem 3, there exist hamiltonian paths $Q_1, Q_2, \dots, Q_{n-2}$ of S_n^n joining $\mathbf{e}$ and some black node $\mathbf{s}_i$ such that (1) $(\mathbf{s}_i)_1 = b_i$ for $1 \leq i \leq n-2$, and (2) $|\{Q_1(i), Q_2(i), \dots, Q_{n-2}(i)\}| = n-2$ for $2 \leq i \leq (n-1)!$. Again, there exist hamiltonian paths $R_1, R_2, \dots, R_{n-2}$ of S_n^{n-1} joining $\mathbf{v}$ and some white node $\mathbf{y}_i$ such that (1) $(\mathbf{t}_i)_1 = b_{i+n-2}$ for $1 \leq i \leq n-2$, and (2) $|\{R_1(i), R_2(i), \dots, R_{n-2}(i)\}| = n-2$ for $2 \leq i \leq (n-1)!$.

By Theorem 2, there exists a path W_i as $\langle \mathbf{x}_1^i, T_1^i, \mathbf{y}_1^i, \mathbf{x}_2^i, T_2^i, \mathbf{y}_2^i, \dots, \mathbf{x}_{n-2}^i, T_{n-2}^i, \mathbf{y}_{n-2}^i \rangle$ joining the white node $\mathbf{x}_1^i$ to the black node $\mathbf{y}_{n-2}^i$ such that $\mathbf{x}_1^i = (\mathbf{s}_i)^n$, $\mathbf{y}_{n-2}^i = (\mathbf{t}_i)^n$, and T_j^i is a hamiltonian path of $S_n^{b_{i+j-1}}$ joining $\mathbf{x}_j^i$ to $\mathbf{y}_j^i$ for $1 \leq j \leq n-1$, and $1 \leq i \leq n-2$. Hence $\{P_1, P_2, \dots, P_{n-2}\}$ form the desired paths. See Figure 2 for illustration the case $n = 5$.

Suppose that $\mathbf{u} = \mathbf{e}$ and $\mathbf{v} = (\mathbf{e})^n$. Let $P_1, P_2, \dots, P_s$ be a set of mutually independent hamiltonian path between $\mathbf{u}$ and $\mathbf{v}$. Obviously, $\{P_i(2) | 1 \leq i \leq s\} \subseteq \{(\mathbf{u})^i | 2 \leq i \leq n\}$. Thus $s \leq n-2$. Thus, the value $(n-2)$ is tight. $\square$

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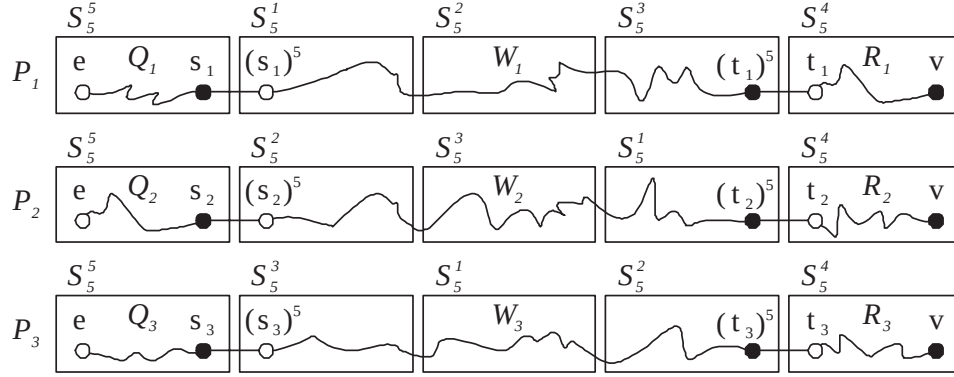


Figure 2: Illustration for 3 mutually independent hamiltonian paths on S_5 .

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